

Black formula - Exercise

Black formula is an analytical formula that gives the price of a European call (or put) option on a forward. The payoff of such a European call option on a forward is:

$$(F_T - K)^+$$

where F_T is the forward price at maturity T and K is the strike of the forward contract. The Black formula states that the time-0 value is:

$$C_0 = P(0, T) (F_0 N(d_1) - K N(d_2))$$

with

$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

- $P(0, T)$ is the discount factor from T to 0 (or more generally the time-0 price of a zero-coupon bond paying 1 at time T)
- F_0 is the current forward price, σ is the volatility of the forward
- $N(\cdot)$ is the standard normal cumulative distribution function.

Exercise: reformulate a call option on a forward as an exchange option and use Margrabe formula to derive Black formula.

Step 1: Reformulate as an Exchange option

↳ The payoff of the call on ~~call~~ forward is :

$$C(T) = \max(F_T - K, 0)$$

We define two assets :

$$\text{Asset 1: } S_1(T) = F_T$$

$$\text{Asset 2: } S_2(T) = K$$

$$\Rightarrow C(T) = \max(S_1(T) - S_2(T), 0)$$

Step 2: $t = 0$

$$S_1(0) = P(0, T) F_0$$

$$S_2(0) = P(0, T) K$$

Since $S_2(T) = K$ is deterministic
we have $\sigma_2 = 0$

$$\Rightarrow \Sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho} = \sigma_1 = \sigma$$

Step 3: applying Margrabe formula

$$C(0) = S_1(0) W(d_1) - S_2(0) W(d_2)$$

$$\hookrightarrow d_1 = \frac{1}{\sigma\sqrt{T}} \left(\frac{1}{2} \sigma^2 T + \ln(F_0/K) \right)$$

$$\hookrightarrow d_2 = d_1 - \sigma\sqrt{T}$$

$$\Rightarrow C(0) = P(0, T) \left(F_0 W(d_1) - K W(d_2) \right)$$

~~4~~